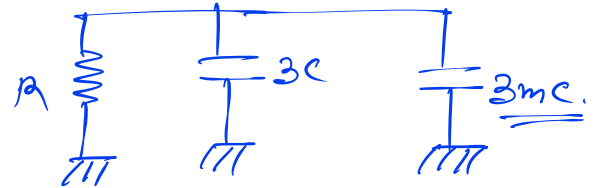


Question 1

Estimate t_{pd} for a unit inverter driving m identical unit inverters. If the driver is w times unit size, what is the delay? If a unit transistor has R=10 kΩ and C=0.1 fF in a 65 nm process, compute the delay, in picoseconds, of the inverter with a fanout of h=4.

$$t_{pd} = (2C + 3mC)R$$

$$= (3 + 3m)RC.$$



$$t_{pd} = (3wC + 3mC) \frac{R}{w} =$$

$$\underline{(3 + 3h)RC} \rightarrow h = \frac{m}{w}$$

Question 2

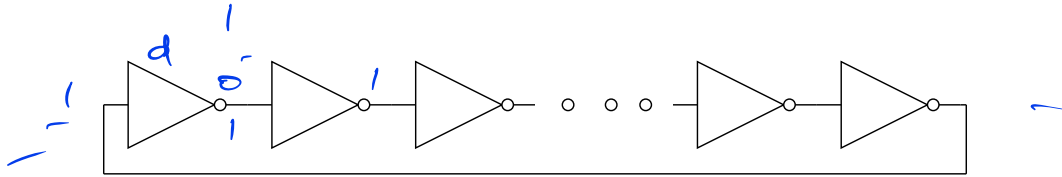
Use the linear delay model to estimate the delay of the fanout-of-4 (FO4) inverter. Assume the inverter is constructed in a 65 nm process with $\tau=3$ ps.

$$d = p + gh = 1 + 1 \times 4 = 1 + 4 = 5$$

$$5\tau = 5 \times 3 = 15 \text{ ps.}$$

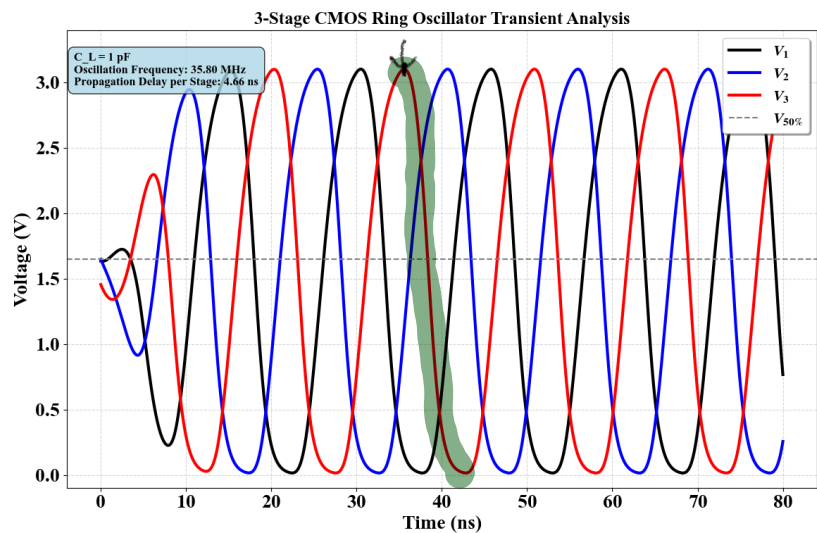
Question 3

A ring oscillator is constructed from an odd number of inverters, as shown in Figure. Estimate the frequency of an N-stage ring oscillator. What will be the frequency of 31 stage ring oscillator if $\tau=3$ ps in a 65 nm process?



$d = \text{delay of single inverter.}$

Total delay
= Nd .



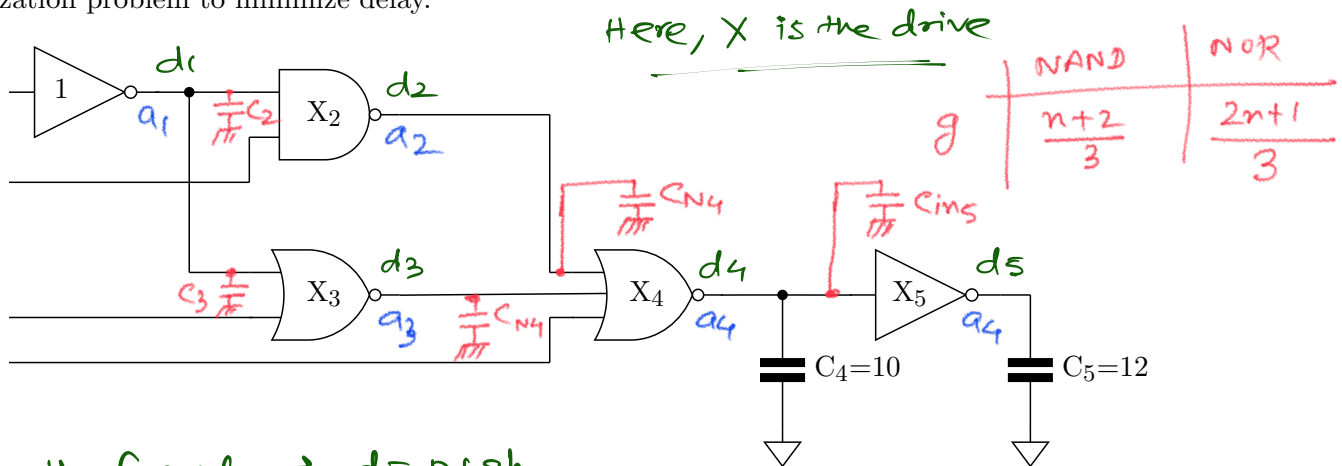
$$f = \frac{1}{T} = \frac{1}{2 \cdot N \cdot d} = \frac{1}{2 \times 31 \times 2 \cdot \tau} = \frac{1}{123 \tau}$$

$$= \frac{1}{123 \times 3 \times 10^{-12}} = 2.71 \text{ GHz}$$

$$d = p + g \cdot h$$

Question 4

The circuit in Figure has nonuniform branching, reconvergent fanout, and a wire load in the middle of the path, all of which stymie back-of-the-envelope application of Logical Effort. The wire load is given in the same units as the gate capacitances (i.e., multiples of the capacitance of a unit inverter). Assume the inputs arrive at time 0. Write an expression for the arrival time of the output as a function of the gate drives. Form the optimization problem to minimize delay.



We use the formula $\rightarrow d = p + gh$.

$$\rightarrow d_1 = p + gh = 1 + 1X \frac{C_{out}}{C_{in}}$$

$$= 1 + 1X \frac{C_2 + C_3}{C_{in}}$$

$$= 1 + \frac{g_2 X_2 + g_3 X_3}{C_{in}}$$

$$= 1 + \frac{\frac{4}{3} X_2 + \frac{5}{3} X_3}{1}$$

$$d_1 = 1 + \frac{4}{3} X_2 + \frac{5}{3} X_3$$

$$\text{drive } X = \frac{C_{in}}{g}$$

$$\therefore C_{in} = gX$$

* Here, C_{out} and C_{in} are represented in a normalized scale, w.r.t. unit inverter capacitance

Actual $\therefore C_{in}$ for inverter $\rightarrow 3C$.

in normalized scale $\rightarrow \frac{3C}{3C} = 1$.

$$d_2 = p_2 + g_2 h_2 = 2 + \frac{4}{3} \cdot \frac{C_{N4}}{C_{in2}} \quad \left[\because \text{2 input NAND gate } p=2, g=\frac{4}{3} \right]$$

$$= 2 + \frac{4}{3} \cdot \frac{g_4 X_4}{g_2 X_2} = 2 + \frac{4}{3} \cdot \frac{\frac{7}{3} X_4}{\frac{4}{3} X_2} = 2 + \frac{7}{3} \frac{X_4}{X_2}$$

$$d_2 = 2 + \frac{7}{3} \frac{X_4}{X_2}$$

$$d_3 = p_3 + g_3 h_3 = 2 + \frac{5}{3} \cdot \frac{C_{N4}}{C_{in3}} = 2 + \frac{5}{3} \cdot \frac{g_4 x_4}{g_3 x_3}$$

$$= 2 + \frac{5}{3} \cdot \frac{\frac{7}{3} x_4}{\frac{5}{3} x_3} = 2 + \frac{7}{4} \frac{x_4}{x_3}$$

$$d_3 = 2 + \frac{7}{4} \frac{x_4}{x_3}$$

$$d_4 = p_4 + g_4 h_4 = 3 + \frac{7}{3} \frac{C_4 + C_{in5}}{C_{in4}} = 3 + \frac{7}{3} \cdot \frac{C_4 + g_5 x_5}{g_4 x_4}$$

$$= 3 + \frac{7}{3} \cdot \frac{10 + x_5}{\frac{7}{3} x_4} = 3 + \frac{10}{x_4} + \frac{x_5}{x_4}$$

$$d_4 = 3 + \frac{10}{x_4} + \frac{x_5}{x_4}$$

$$d_5 = p_5 + g_5 h_5 = 1 + 1 \times \frac{C_5}{C_{in5}} = 1 + \frac{12}{g_5 x_5} = 1 + \frac{12}{x_5}$$

$$d_5 = 1 + \frac{12}{x_5}$$

Arrival time

$$\rightarrow a_1 = d_1$$

$$\rightarrow a_2 = a_1 + d_2$$

$$\rightarrow a_3 = a_1 + d_3$$

$$a_4 = \max\{a_2; a_3\} + d_4$$

$$= \max\{a_1 + d_2; a_1 + d_3\} + d_4$$

$$= \max\{d_1 + d_2; d_1 + d_3\} + d_4$$

$$\rightarrow a_4 = d_1 + \max\{d_2; d_3\} + d_4$$

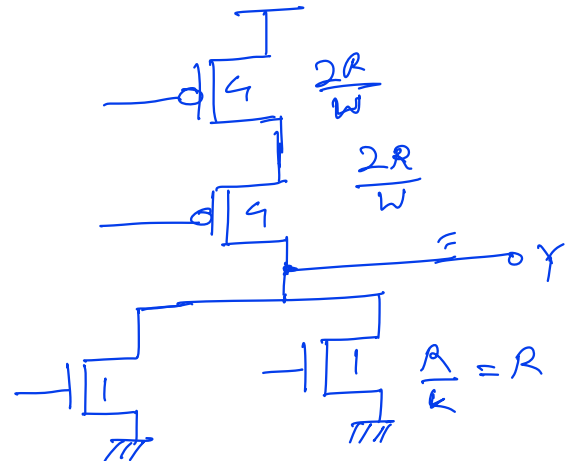
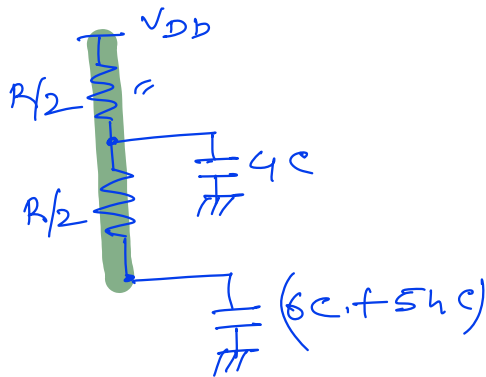
$$a_5 = a_4 + d_5$$

$$\rightarrow a_5 = d_1 + \max\{d_2; d_3\} + d_4$$

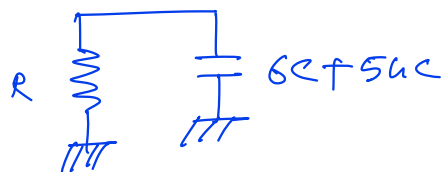
Thus, we have to minimize a_5 with d_1, d_2, d_3, d_4, d_5 constraint.

Question 5

Sketch a 2-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance.

 τ_{pdr} 

$$\tau_{pdr} = 4C(R/2) + (6C + 5hC)R = \underline{\underline{8RC + 5hRC}}$$

 τ_{pdf} 

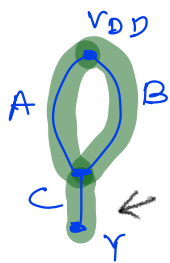
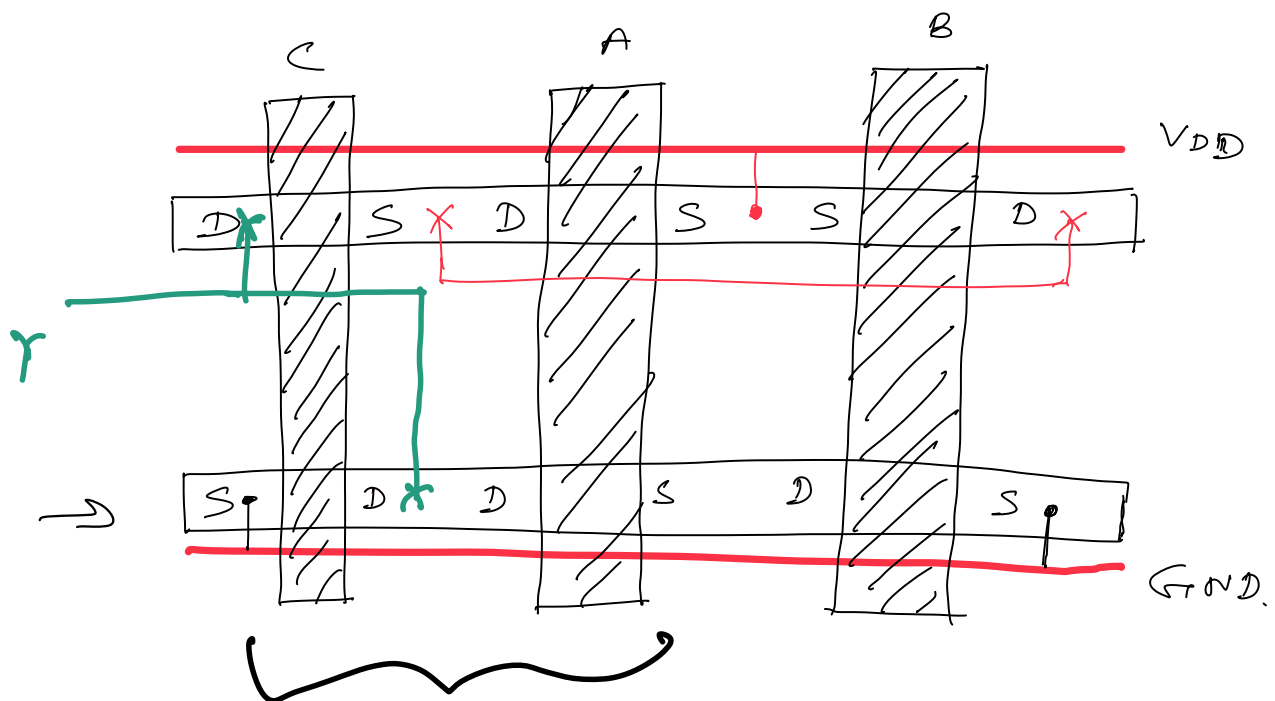
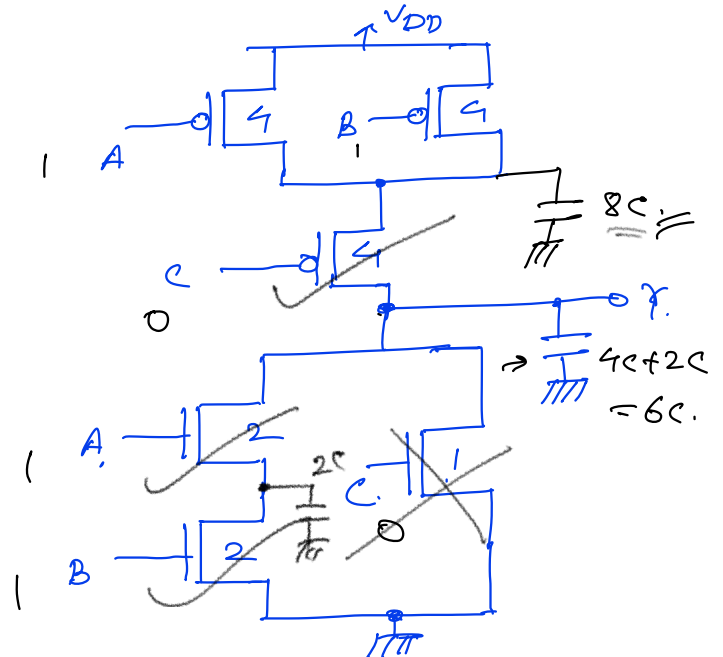
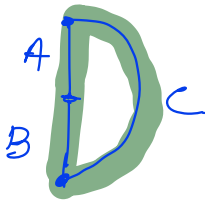
$$\tau = (6C + 5hC)R$$

Question 6 AOI2.1

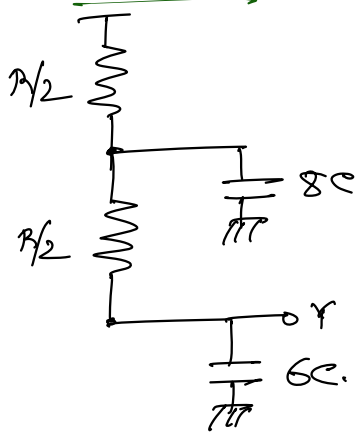
Find the rising and falling propagation delays of an unloaded AND-OR-INVERT gate using the Elmore delay model. Estimate the diffusion capacitance based on a stick diagram of the layout.

Soln:

$$\text{AOI2.1} \Rightarrow Y = \overline{AB + C}$$

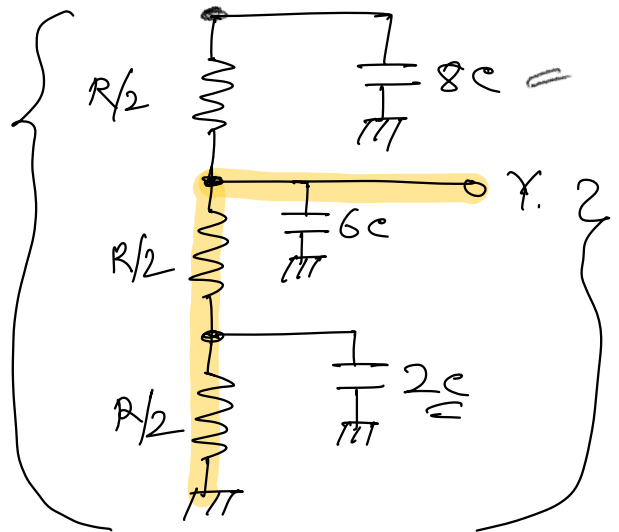
PUNC-A-BPDN

done in class



$$\tau = 8C \cdot \frac{R}{2} + 6C \times \left(\frac{R}{2} + \frac{R}{2} \right) = 10RC.$$

falling



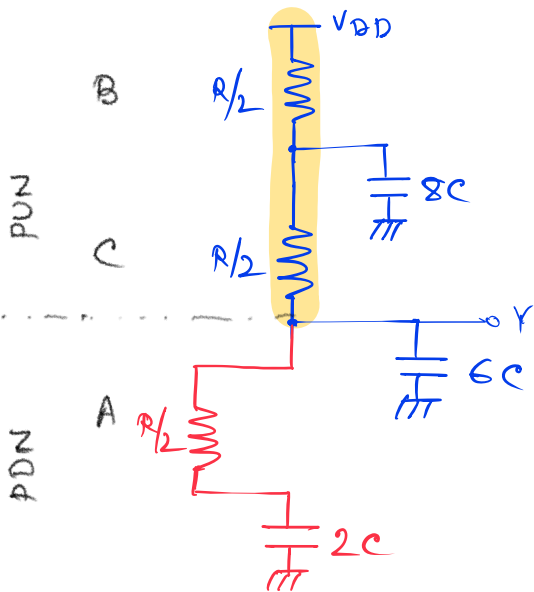
$$= \frac{2C \cdot R}{2} + 6C \cdot R + 8C \cdot R = 15RC$$

Actual (Suggested by Urjit)

For propagation rising delay; Either of A or B should be on for PUN.

Say A=1, B=0

Supply $\rightarrow V_{DD}$; Target node $\rightarrow Y$



Elmore delay =

$$= 8C \times \frac{R}{2} + 6C \times \left(\frac{R}{2} + \frac{R}{2} \right) + 2C \times \left(\frac{R}{2} + \frac{R}{2} \right)$$

$$= 4RC + 6RC + CR$$

$$= \underline{\underline{11RC}}$$

Question 7

Find the worst-case Elmore parasitic delay of an n-input NOR gate.

Solⁿ: At starting, we are not sure about is it propagation delay for rising or falling.
So, we will calculate both and compare.

Rising propagation delay for rising; can be if all PMOS transistors are simultaneously turned on from Off state.

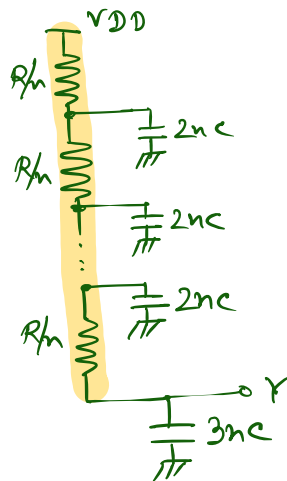
Target Node $\rightarrow Y$

Ref. node $\rightarrow VDD$

Elmore delay

$$= 3nC(R/n + \dots + n \text{ times})$$

$$+ 2nC \sum_{i=1}^{n-1} i \cdot \frac{R}{n}$$



$$= 3nC \left(\frac{R}{n} \cdot n \right) + 2nC \frac{R}{n} \sum_{i=1}^{n-1} i = 3nRC + 2RC \frac{n-1}{2} (1+n-1)$$

$$= 3nRC + n(n-1)RC = n^2RC + 2nRC = \underline{(n^2 + 2n)RC} = \tau_{pdr}$$

falling The slowest discharging can happen only if the top PMOS (i.e. with input n) is off.

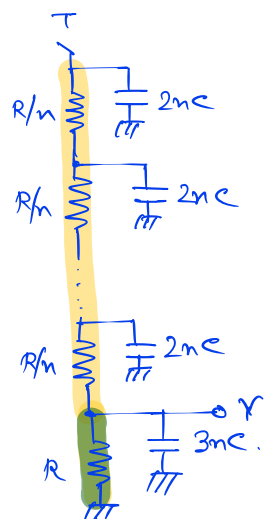
Thus, Corresponding NMOS must be on for that input.

Target Node $\rightarrow Y$ | Elmore delay = $3nCR + 2nC \sum_{i=1}^{n-1} R$

$$= 3nCR + 2nC(n-1)R$$

$$= 2n^2RC + nRC$$

$$\tau_{pdf} = \underline{(2n^2 + n)RC}$$



$$\tau_{pdf} - \tau_{pdr} = (2n^2 + n - n^2 - 2n) RC = (n^2 - n) RC.$$

$$\therefore n > 1 \text{ thus } n^2 - n > 0. \quad \therefore \tau_{pdf} > \tau_{pdr}.$$

Thus, worst case Elmore parasitic delay is τ_{pdf} .

Question 8

Let a 4x inverter have transistors four times as wide as those of a unit inverter. If a unit inverter has three units of input capacitance and parasitic delay of ρ_{inv} , what is the input capacitance of a 4x inverter? What is the logical effort? What is the parasitic delay?

$$\text{input capacitance} = 4 \times 3C = 12C.$$

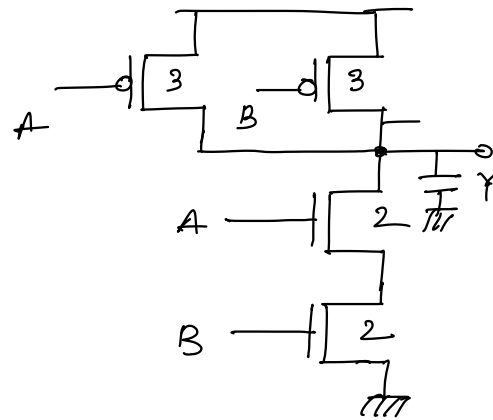
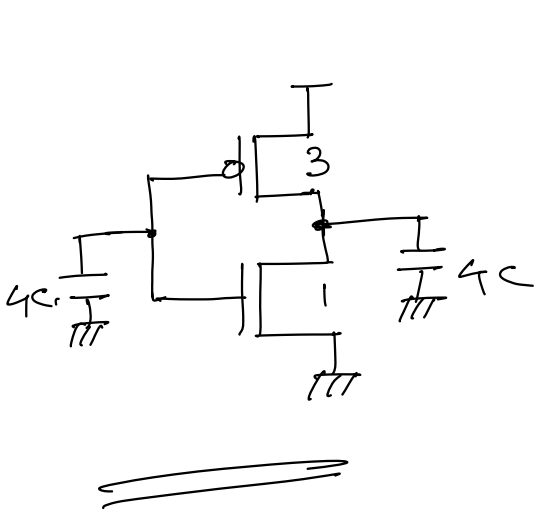
$$g = \frac{12C}{12C} = 1$$

$$p = 12C \times \frac{R}{4} = 3RC = \underline{\underline{3\rho_{inv}}}$$

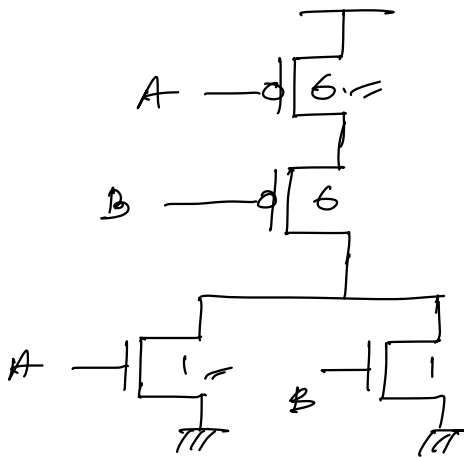
Question 9

Consider a process in which pMOS transistors have three times the effective resistance as nMOS transistors.

Calculate the logical efforts of a 2-input NAND gate and a 2-input NOR gate if they are designed with equal rising and falling delays.



$$g = \frac{5C}{4C}$$



$$g = \frac{7}{4}$$

